

HGSTA: Leveraging Hypergraph Computing for Effective Collaborative Perception Feature Fusion

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Abstract—Collaborative perception is a vital approach for autonomous systems to understand complex environments by sharing information among multiple agents. However, existing methods often struggle with spatiotemporal misalignment and inefficient feature fusion under dynamic, uncertain conditions. To address these challenges, we propose HGSTA (HyperGraph Spatial-Temporal Awareness), a novel framework that integrates hypergraph computing for multi-scale spatiotemporal feature fusion. By modeling high-order relationships among agents' data through a hypergraph, HGSTA effectively aligns and combines complementary features from different viewpoints. Extensive experiments on three benchmark datasets demonstrate that HGSTA significantly outperforms current state-of-the-art methods, validating the advantages of our hypergraph-based collaborative perception approach.

Index Terms—collaborative perception, hypergraph, robustness, heterogeneous data

I. INTRODUCTION

Collaborative perception in autonomous driving and robotics enables multiple agents (e.g., vehicles or robots) to share sensor data or intermediate features, thereby improving perception of complex, occluded, and dynamic environments [1], [2]. By leveraging data from diverse viewpoints, this paradigm helps mitigate limitations of single-agent perception, such as sensor occlusions and limited fields of view [3], [4].

Collaborative perception methods are typically categorized into early, intermediate, and late fusion [5], [6]. While early fusion transmits raw sensory data and late fusion shares high-level predictions, both suffer from trade-offs between bandwidth consumption and information loss. Intermediate fusion, which exchanges semantic features extracted from raw data, offers a favorable balance between communication efficiency and perception quality [7], [8].

However, intermediate fusion still faces challenges due to spatiotemporal misalignments—caused by sensor heterogeneity, latency, and localization errors—which degrade the accuracy of fused features [8], [9]. Most prior works focus on pairwise relationships or heuristic communication strategies, neglecting higher-order dependencies across agents, which are crucial for robust feature aggregation in real-world settings.

To address these limitations, we propose to exploit hypergraph structures, which generalize traditional graphs by allowing each hyperedge to connect multiple nodes [10]. This

enables modeling complex, multi-agent interactions and capturing joint spatiotemporal-semantic correlations. We develop a novel Hypergraph-based Spatiotemporal Attention (HGSTA) framework that performs multi-scale feature fusion, addressing both spatial inconsistencies and latent temporal shifts by learning adaptive weights and structural dependencies among agents.

Our key contributions are summarized as follows:

- We propose a hypergraph construction method tailored for collaborative perception, effectively modeling high-order relationships among agents and semantic features.
- We design a multi-scale fusion framework based on hypergraph computing, capable of aligning spatial, temporal, and semantic information across agents under noisy and dynamic conditions.
- We demonstrate that integrating high-order features via hypergraph-based fusion significantly improves perception robustness and accuracy, validated through experiments on multiple collaborative driving benchmarks.

II. RELATED WORK

A. Collaborative Perception in Autonomous Systems

Collaborative perception has received increasing attention as a means to enhance situational awareness by allowing multiple autonomous agents to share and fuse sensory information [11]–[13]. Early works such as Cooper [3] illustrated the effectiveness of feature sharing in V2X systems. Later approaches introduced more selective strategies for collaboration. When2com [9] proposed learning when to communicate based on temporal dynamics, while Where2comm [8] emphasized spatially guided feature exchange. Other recent frameworks leverage multi-module architectures [14] or multi-modal data fusion [15], [16] to improve perception efficiency.

Despite promising results, many approaches still rely on pairwise or local interactions among agents [6], [7]. They often struggle with real-time constraints, dynamic agent configurations, and data misalignment due to sensor or communication delays [17]. Robust fusion under these conditions remains an open challenge.

B. Hypergraph Computation and Its Application

Hypergraphs generalize traditional graphs by allowing hyperedges to connect multiple nodes, making them suitable for modeling high-order interactions in data [10]. In computer vision and multi-agent systems, hypergraph-based methods have been used for image correspondence [18], collaborative classification [19], and agent coordination [20]. Wu et al. [19] demonstrated that hypergraph convolution networks (HGCNs) can effectively propagate information in settings with complex interdependencies.

In autonomous systems, hypergraph formulations remain underexplored. Although recent work such as HCoN [21] provides insights into stability and robustness, few studies integrate hypergraph modeling into real-time, multi-agent perception frameworks. Furthermore, challenges such as computational overhead and scalability persist in high-dimensional, dynamic environments.

Our work aims to bridge this gap by introducing a hypergraph-based feature fusion method tailored for collaborative perception, demonstrating that high-order representations can significantly improve alignment and robustness in noisy, real-world scenarios.

III. METHODS

A. Framework Overview

HGSTA (HyperGraph Spatiotemporal Attention) is a unified framework designed to improve collaborative perception by addressing two critical challenges: insufficient modeling of high-order relations and spatial misalignment across agents. As shown in Fig. 1, the framework comprises three stages: **Encoding**, **Fusion**, and **Decoding**.

Encoding: Each agent i captures raw sensory input $\mathbf{X}_i$ (e.g., LiDAR point cloud) and encodes it into semantic features using a shared backbone:

$$\mathbf{F}_i = f_{\text{encode}}(\mathbf{X}_i), \quad \mathbf{F}_i \in \mathbb{R}^{N_i \times c} \quad (1)$$

This step ensures all agents project their observations into a unified feature space. The features from all M agents are collected as:

$$\mathcal{F} = \{\mathbf{F}_1, \mathbf{F}_2, \dots, \mathbf{F}_M\} \quad (2)$$

Fusion: To effectively integrate information across agents and scales, we propose a multi-scale high-order fusion strategy. At each level j , semantic features are extracted and fused:

$$\begin{aligned} \mathbf{S}_{\text{level}_j} &= g_{\text{semantic}}(\mathcal{F}), \\ \mathbf{Z}_j &= f_{\text{fuse}}(\mathbf{S}_{\text{level}_j}), \\ \mathbf{Z} &= \text{Concat}(\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_j) \end{aligned} \quad (3)$$

This hierarchical fusion design allows the network to capture both fine-grained and global contextual dependencies from diverse perspectives.

Decoding: The fused representation $\mathbf{Z}$ is processed via classification and regression heads to generate detection results:

$$\hat{y}_{\text{class}} = \text{Softmax}(W_c \mathbf{Z}), \quad \hat{y}_{\text{reg}} = W_r \mathbf{Z} \quad (4)$$

where the heads estimate category scores and 3D bounding box parameters, respectively.

B. Training Objective

The model is optimized by a joint loss function:

$$\mathcal{L} = \mathcal{L}_{\text{class}} + \lambda \mathcal{L}_{\text{reg}} \quad (5)$$

where $\mathcal{L}_{\text{class}}$ and $\mathcal{L}_{\text{reg}}$ denote classification and regression loss terms. To evaluate perception quality, we adopt Average Precision (AP) as the metric, and define the final optimization objective as:

$$\max \text{AP} = f_{\text{eva}}(f_{\text{dec}}(f_{\text{hyperfuse}}(f_{\text{encode}}(\mathbf{X})), \mathbf{Y}_i)) \quad (6)$$

ensuring end-to-end improvement from raw features to detection performance.

C. Hypergraph-based Semantic Fusion

Traditional fusion methods often rely on simple concatenation or pairwise attention, which are limited in modeling group-wise interactions. To overcome this, we propose HyperC2Net, a hypergraph-based module that captures latent high-order relations across multiple scales and agents.

We begin by extracting features from multiple layers of a shared backbone [22]. These multi-scale features— $\{\mathbf{B}_{\text{low}}, \mathbf{B}_{\text{mid}}, \mathbf{B}_{\text{high}}\}$ —are aligned to a common resolution through:

$$\mathbf{F} = C(U(\mathbf{B}_{\text{low}}), \mathbf{B}_{\text{mid}}, D(\mathbf{B}_{\text{high}})) \quad (7)$$

where $U(\cdot)$ and $D(\cdot)$ are up/downsampling operations, and $C(\cdot)$ denotes channel-wise concatenation.

To construct a hypergraph $G = (V, E)$, we adopt SPD-HC [23], which measures relationships based on set-pair distances. For each node pair, we define:

$$u = \frac{S_{\text{count}}}{c} + \frac{F_{\text{count}}}{2c} - \frac{P_{\text{count}}}{c}, \quad \text{distance} = 1 - u \quad (8)$$

categorizing node similarity into same, uncertain, and opposite attributes. Edges are formed when distances fall below a threshold, enabling flexible connectivity across semantic neighborhoods.

We then perform message passing on the hypergraph. This occurs in two stages:

$$x_e = \frac{1}{|\mathcal{N}_v(e)|} \sum_{v \in \mathcal{N}_v(e)} x_v \circ \Theta, \quad x'_v = x_v + \frac{1}{|\mathcal{N}_e(v)|} \sum_{e \in \mathcal{N}_e(v)} x_e, \quad (9)$$

which correspond to aggregation from nodes to hyperedges and redistribution back to nodes. Here, Θ is a learnable transformation.

For computational efficiency, this process can be expressed in matrix form:

$$\text{HyperConv}(X, H) = X + D_v^{-1} H D_e^{-1} H^\top X \Theta. \quad (10)$$

This enables efficient and scalable group-level feature refinement across agents and resolutions.

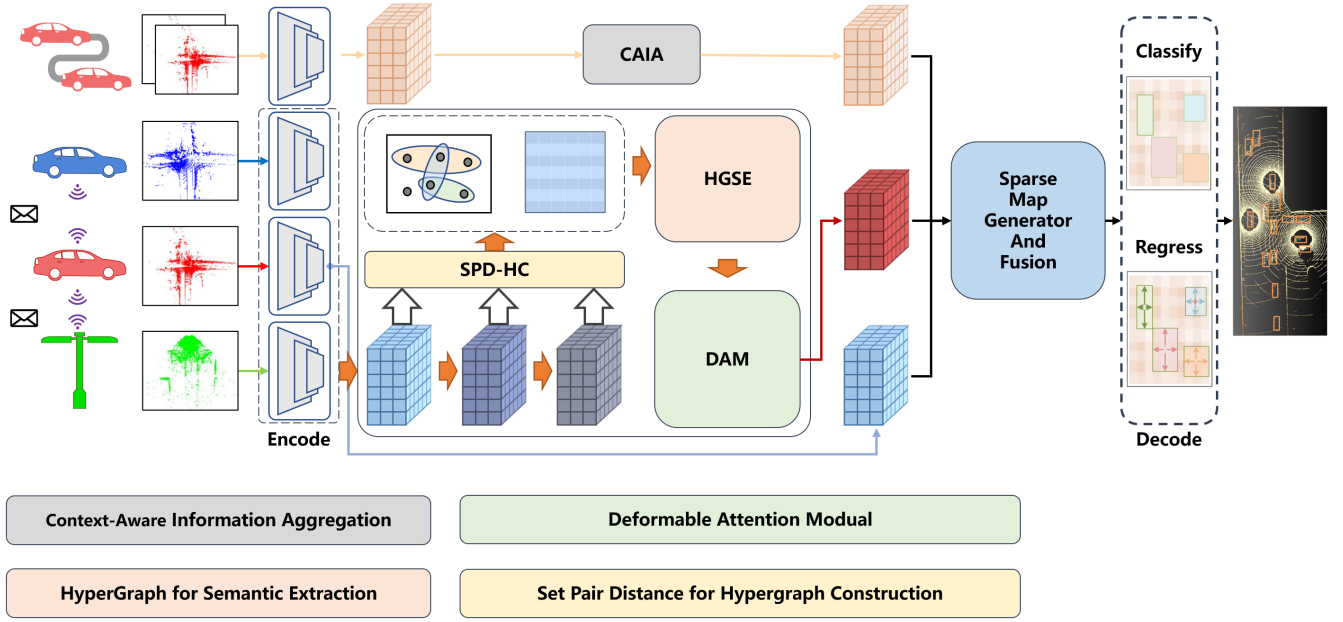


Fig. 1. Overview of the HGSTA framework. The pipeline includes shared feature encoding, hypergraph-based multi-scale fusion, deformable attention refinement, and final detection.

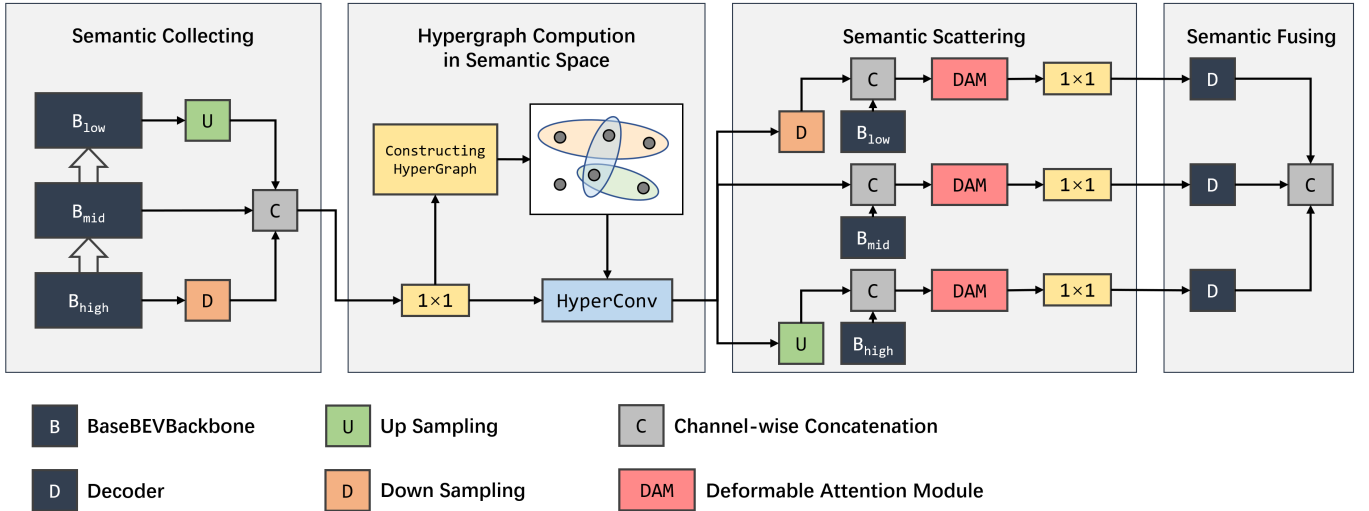


Fig. 2. Architecture of HyperC2Net. It aligns multi-scale features across agents, constructs a hypergraph via SPD-HC, and performs high-order message propagation via HyperGraph Convolution.

D. Deformable Attention Module Refinement

Even after semantic fusion, feature misalignment—caused by localization errors or asynchronous transmission—can hinder perception. To mitigate this, we introduce a Deformable Attention Module (DAM) [24], which adaptively samples relevant context via learned spatial offsets.

Given a query point q , DAM predicts M offsets Δq_m , and samples local features from $q + \Delta q_m$. Attention is computed across heads as:

$$\text{DAM}(q) = \sum_{a=1}^A W_a \left[\sum_{k=1}^K \sum_{m=1}^M \phi(W_b F(q)) F(q + \Delta q_m) \right] \quad (11)$$

where ϕ is the softmax function. The output is fused back via residual connection:

$$Z_{i,l}(q) = F(q) + \text{DAM}(q) \quad (12)$$

DAM complements hypergraph fusion by focusing on spatial alignment. It helps the model learn adaptive sampling

regions that vary with resolution and viewpoint, effectively reducing the impact of ego-pose uncertainty. We further inject hypergraph-derived semantics into DAM inputs, enabling it to align not only geometric structure but also semantic intent.

In summary, the combination of high-order hypergraph reasoning and deformable spatial refinement equips HGSTA with both global awareness and local adaptability, advancing collaborative perception in real-world V2X scenarios.

IV. EXPERIMENTS

We evaluate the proposed HGSTA framework on three benchmark datasets for collaborative 3D object detection. Our experiments aim to verify: (1) performance improvements over SOTA methods; (2) robustness under real-world disturbances (e.g., noise and delays); and (3) the contribution of each core module through ablation studies.

A. Datasets and Evaluation Metrics

Datasets. We conduct experiments on OPV2V [5], V2XSet [6], and V2V4Real [7]:

- *OPV2V* is a simulated V2V dataset with 11k+ multi-agent frames and annotated LiDAR and camera data.
- *V2XSet* extends to V2X scenarios with 73 traffic scenes and 11.4k point cloud frames.
- *V2V4Real* is the first large-scale real-world V2V dataset, comprising 20k LiDAR frames, 40k images, and 240k 3D bounding boxes.

Metrics. We follow [6] and report 3D object detection performance using Average Precision (AP) under IoU thresholds of 0.5 and 0.7. To evaluate communication cost, we use log-scale message size in bytes [8].

B. Implementation Details

All models are implemented in PyTorch [25] and trained on two RTX 4090 GPUs for 30 epochs per dataset. We use the Adam optimizer [26] with an initial learning rate of 2×10^{-3} , decayed by 0.1 after epoch 15. To stabilize early training, a linear warm-up strategy is applied over the first 5 epochs, followed by cosine annealing. Batch size is set to 16.

The encoder voxel resolution is 0.4 m. For DAM, we set 5 keypoints and 8 attention heads. To simulate real-world noise, we inject Gaussian localization errors (std=0.2 m, 0.2°) and apply random augmentations including $\pm 10^\circ$ yaw and small translations. These settings help assess robustness under imperfect spatial synchronization.

C. Quantitative Results

Table I reports 3D detection results. Compared to single-agent baselines and state-of-the-art fusion models like V2X-ViT [6], V2VNet [27], and CoBEVT [28], HGSTA consistently achieves the best performance across all benchmarks. Notably, on OPV2V and V2XSet, HGSTA surpasses the previous best by 5.06% and 3.28% AP@0.7 respectively, validating the benefit of our hypergraph-based fusion strategy. As shown in Fig. 3, HGSTA detects occluded vehicles missed by baselines. Our predictions better align with ground truth, validating feature consistency.

TABLE I
PERFORMANCE COMPARISON ACROSS DATASETS

Dataset	Methods	IoU 0.5	IoU 0.7
OPV2V	No fusion	68.71	48.66
	V2X-ViT	84.76	68.92
	V2VNet	77.58	62.27
	Where2comm	80.55	61.92
	CoBEVT	81.32	63.19
	CoBEVFlow	83.06	71.08
	MRCNet	82.59	70.45
	HGSTA	86.14	73.98
V2XSet	No fusion	60.60	40.22
	V2X-ViT	83.59	61.44
	V2VNet	79.11	49.28
	Where2comm	81.98	53.43
	CoBEVT	81.12	54.30
	CoBEVFlow	82.13	62.42
	MRCNet	82.97	63.31
	HGSTA	84.10	64.72
V2V4Real	No fusion	39.78	22.02
	V2X-ViT	55.35	28.72
	V2VNet	57.16	26.73
	Where2comm	58.37	31.49
	CoBEVT	58.20	28.72
	CoBEVFlow	59.34	33.73
	MRCNet	59.47	35.49
	HGSTA	60.19	37.89

1) *Robustness to Localization Error:* Collaborative methods are vulnerable to spatial misalignment. We simulate localization noise using Gaussian perturbation ($\sigma_{xyz} \in [0, 0.5]$ m). As shown in Fig. 4, HGSTA maintains stable performance under increasing noise, demonstrating superior resilience compared to existing methods. This is attributed to the hypergraph-based feature alignment, which captures high-order context beyond absolute coordinates.

2) *Robustness to Transmission Delay:* We further simulate communication delays up to 200 ms. Fig. 5 shows that HGSTA significantly outperforms SOTA methods under various delay levels. This robustness stems from the deformable attention mechanism, which adaptively samples context points regardless of misaligned timestamps.

D. Ablation Study

We conduct ablation studies to isolate the contributions of HGSE (HyperGraph Semantic Extraction) and DAM. Table II shows their individual and combined impacts. Introducing HGSE improves semantic representation via group-level reasoning, while DAM enables dynamic feature alignment. Their combination leads to consistent performance gains across all datasets.

In addition to accuracy, we observe that the full HGSTA model converges faster and exhibits lower variance during training, indicating better optimization stability.

E. Hypergraph Construction Analysis

To evaluate our hypergraph construction strategy, we compare SPD-HC (Set Pair Distance) with Euclidean and cosine distance-based graphs. Table III shows that SPD consistently yields the best performance, confirming its strength in capturing complex relational structures beyond geometric proximity.

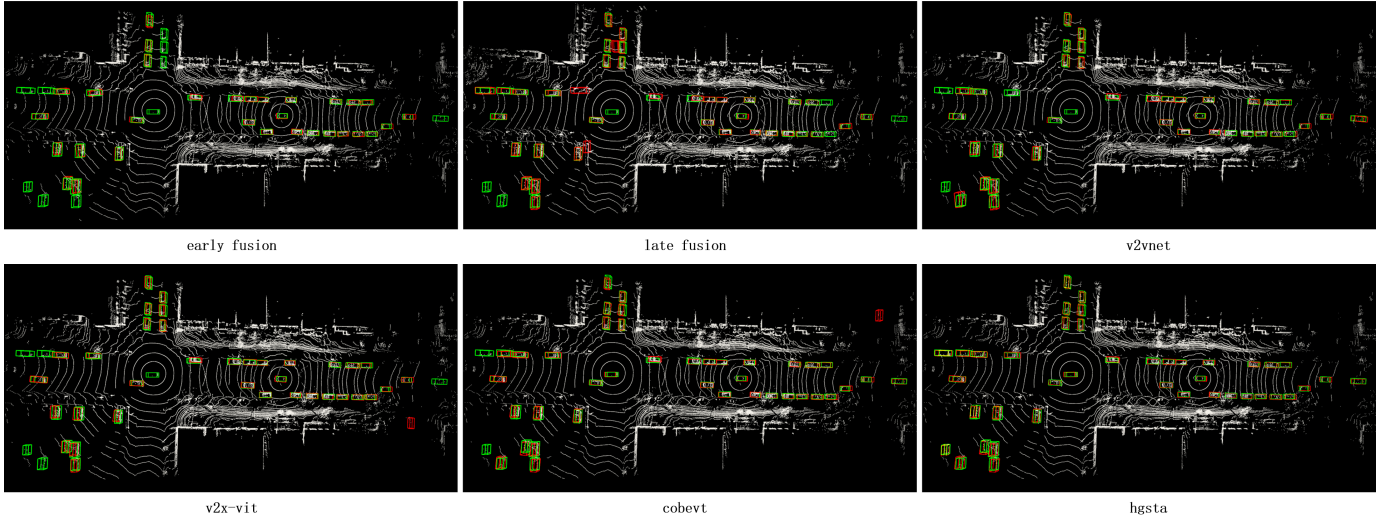


Fig. 3. Qualitative comparison of detection results. Green boxes: ground truth; red: predictions. From top to bottom: Early Fusion, Late Fusion, V2VNet, V2X-ViT, CoBEVT, HGSTA.

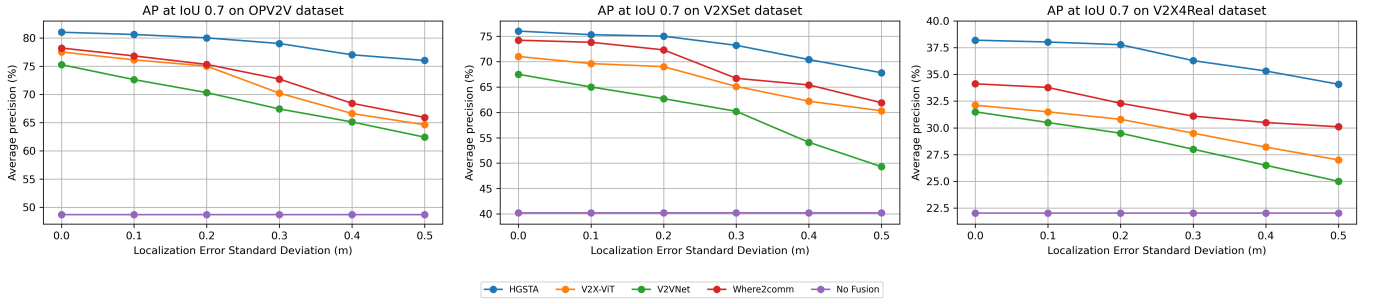


Fig. 4. Robustness to localization errors: HGSTA shows less accuracy degradation under spatial noise.

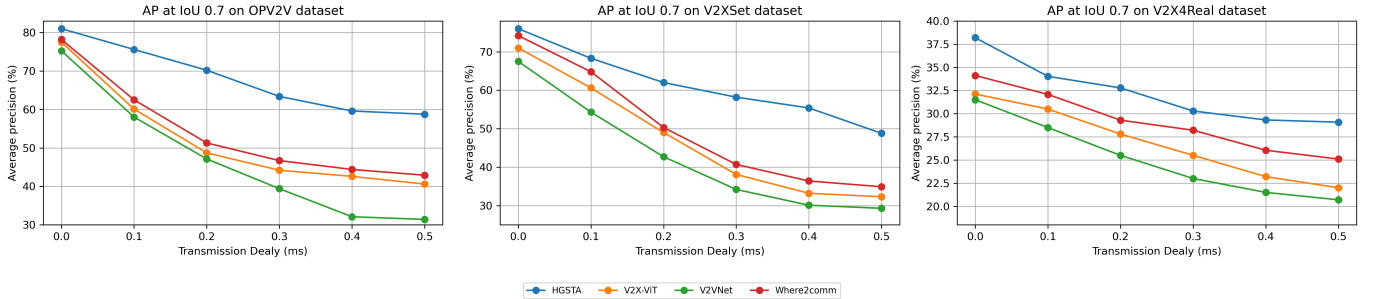


Fig. 5. Performance under different transmission delays. HGSTA retains stable detection accuracy.

V. CONCLUSION

In this paper, we presented HGSTA, a hypergraph-based framework for effective collaborative perception. HGSTA introduces a novel way to model and fuse multi-agent features by capturing high-order spatial and temporal relationships through hypergraph structures. Our multi-scale hypergraph fusion approach addresses the misalignment and limited pairwise modeling of previous methods, resulting in more robust and accurate perception in multi-agent scenarios. We demonstrated that HGSTA achieves state-of-the-art performance on chal-

lenging autonomous driving datasets, significantly outperforming existing collaborative perception techniques. These results illustrate the potential of hypergraph computing in enhancing perception for connected autonomous systems and beyond.

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TABLE II
IMPACT OF DIFFERENT MODULES ON PERFORMANCE

Dataset	HGSE	SPD-HC	DAM	IoU 0.5	IoU 0.7
OPV2V	✓			77.69	62.45
	✓	✓		82.31	68.54
	✓	✓	✓	86.14	73.98
V2XSet	✓			60.60	40.22
	✓	✓		81.98	53.43
	✓	✓	✓	84.10	64.72
V2V4Real	✓			39.78	22.02
	✓	✓		55.46	34.49
	✓	✓	✓	60.19	37.89

TABLE III
IMPACT OF GRAPH CONSTRUCTION METHODS

Dataset	Method	IoU 0.5	IoU 0.7
OPV2V	Euclidean	77.69	65.34
	Cosine	82.31	70.69
	Set-Pair	86.14	73.98
V2XSet	Euclidean	77.53	55.44
	Cosine	79.45	58.83
	Set-Pair	84.10	64.72
V2V4Real	Euclidean	56.44	30.45
	Cosine	57.61	32.83
	Set-Pair	60.19	37.89

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